Predictive Analytics for E-commerce Product Replenishment: A Machine Learning Approach

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***Abstract*—This project applies predictive analytics to forecast customer repurchase timing in e-commerce. Using a dataset of over 1 million transactions and product details, we employed machine learning models, including XGBoost and CatBoost, to predict repurchase windows. Key techniques included feature engineering, temporal analysis, and product grouping using Jaccard similarity. XGBoost achieved the best performance with a final score of 7170. Despite challenges like data imbalance, the study demonstrates the potential of predictive models to optimize inventory management and enhance supply chain efficiency, offering valuable understanding for future applications in e-commerce.**

***Index Terms*—customer segmentation, purchase prediction**

I. Introduction

In the rapidly evolving e-commerce ecosystem, leveraging predictive analytics to enhance customer experience has become the cornerstone of success, enabling businesses to not only understand but also anticipate customer needs with remarkable precision and foresight. Our project focuses on predicting user behavior using transactional data and product catalogs in household and consumable products.

A. Problem Statement

Several data mining algorithms are used to predict whether a customer will make a purchase in the next 4 weeks and, if so, when that purchase will occur. The project aims to enhance the platform's costs and reduced delivery times by optimizing inventory management and supply chain operations.

B. Dataset Information

The dataset consists of transactional data, product features, categorical data with parent category. The dataset is stored in a CSV file format. The transactional table, includes customer ID, product ID, purchase date, and purchase amount over the span of 8 months. The table contains approximately 1,000,000 rows and 4 columns. The product features table consists product id, manufacturer id, 5 product features, and categories which product belongs to. Lastly the categorical data table includes category, and parent category which category belongs.

II. Literature Review

Predictive analytics and machine learning have been extensively applied in the e-commerce domain to understand customer behavior, predict purchases, and enhance operational efficiency. This section reviews recent research and methodologies relevant to the objectives of this study. The methodologies and insights from these studies have influenced our approach to predicting customer repurchase behavior and developing effective feature engineering strategies.

A. Predictive Models in E-commerce

Wang et al. [1](2023) proposed an XGBoost-based user behavior prediction model that demonstrated superior performance in e-commerce applications. By leveraging multi-feature fusion and feature importance analysis, their model achieved an accuracy rate above 95%, outperforming traditional machine learning algorithms. The study highlights the significance of efficient feature engineering in improving computational performance and predictive accuracy.

B. Next-Basket Recommendation Systems

Shao et al. [2](2023) conducted a comprehensive empirical study on Next-Basket Recommendation Systems (NBRs), a problem closely related to next-item recommendation problem we worked on. Their research explores diverse methodologies, including Markov Chains, embedding-based techniques, and Recurrent Neural Networks (RNNs), to model both short-term and long-term customer preferences. The study emphasizes the need for unified evaluation frameworks to standardize comparative analyses across NBR methods.

C. Applications of RFM Modeling and Clustering

The RFM (Recency, Frequency, and Monetary) model, reviewed by Wei et al. [3](2010), has been widely utilized for customer segmentation and direct marketing. This model identifies valuable customers by analyzing transaction patterns, enabling targeted marketing strategies. The integration of RFM analysis into predictive analytics provides a robust foundation for customer-centric approaches in e-commerce.
Chang et al. [4](2007) proposed a model to predict customer purchasing behavior by using a combination of clustering analysis and association rules. This model identifies patterns in customer data to group similar customers and then uses association rules to analyze the purchasing.

These studies collectively demonstrate the potential of machine learning and data mining techniques in advancing predictive analytics for e-commerce applications. The methodologies reviewed form the foundation for the approach adopted in this study.

III. Methodology

A. Dataset Description

There are three central components to the dataset : transaction records, product catalog and category hierarchy information.

1) Transaction Data: The primary dataset contains 1,071,538 transaction records spanning 244 days, which is kept in a CSV file with 4 columns:

- Customer ID
- Product ID
- Purchase date
- Purchase quantity

Transaction data provides a inclusive view of purchasing patterns over an 8-month period, covering purchases made by 46,138 unique customers in 31,756 distinct products.

2) Product Catalog: The product catalog contains comprehensive details about each product:

- Product ID (unique identifier)
- Manufacturer ID
- 5 categorical product attributes
- Category assignments (products can belong to multiple categories or none)

3) Category Hierarchy: The category structure is represented in a hierarchical format 1. The complex relationship between product categories and their parent categories can be observed by visualization in a circular layout. Despite offering insights into the product classification system, this hierarchical information was not included in the final analysis due to it's complexity and limited predictive value.

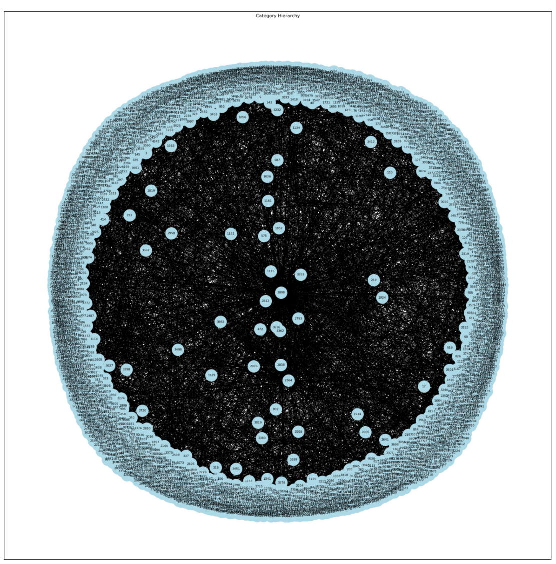

Fig. 1. Categories and their parent categories represented as a graph

B. Data Preprocessing

We used a product grouping strategy using Jaccard similarity to adress the data sparsity problem in our transaction dataset, which consisted of more than 1,000,000 transactions spread across many products and customers. This approach assists in the consolidation of similar products and the developement of more robust purchase patterns. The Jaccard similarity between two products is calculated based on their attributes. Products with a similarity score of 1.0 (identical attributes) are grouped together and assigned a single product ID. A mapping dictionary is used to implement this grouping in the following steps:

1) Each product pair's similarity is computed using their attributes
2) Products with identical attributes are detected
3) For every group a representative product ID is selected
4) Every product in a group are mapped to their representative ID

While preserving the semantic relationships between products, this consolidation strategy successfully reduces the product space. These generated product IDs are used in the following feature engineering steps to create more significant predictive variables.

C. Feature Engineering

To capture the several aspects of the customer's purchase behaviour we developed a extensive feature set from the transaction data. We processed the transaction data by first grouping customers and products to create unique customer-product pairs, then chronologically analyzing their transaction histories. For each transaction date within these groups, we extracted comprehensive features that can be divided into five major categories:

1) Temporal Features: These features capture the timing and patterns of customer purchases, offering insights on buying habits and behaviors. The temporal features include:

a) Purchase History Metrics:

- Days since first and last purchase to measure customer engagement length and recency
- Purchase span days to understand the total active period
- Purchase frequency to quantify buying regularity

b) Interval Analysis:

- Average and median intervals between purchases to understand typical buying cycles
- Standard deviation, shortest, and longest intervals to capture purchase timing variability
- Interval skewness and kurtosis to measure the distribution of purchase timings

c) Calendar-Based Indicators:

- Day of week and month indicators
- Weekend purchase flags
- Month start/end markers for seasonal patterns

2) Purchase Behavior Features: These features measure the direct purchasing actions of customers:

a) Quantity Metrics:

- Last purchase quantity to capture recent purchase behavior
- Total quantity to measure overall volume of purchases
- Purchase count to track frequency of transactions

3) RFM Analysis Features: RFM (Recency, Frequency, Monetary) analysis is a popular method for segmenting customers which uses three key metrics to evaluate customer behavior:

a) Core RFM Metrics:

- Recency: Determines how recently a customer made their last purchase, calculated as the number of days since their last transaction. More recent purchases are indicated by lower values.
- Frequency: Represents how often a customer makes purchases, measured by the total number of transactions. Higher values indicate more frequent buyers.
- Monetary: Reflects the total monetary value of a customer's purchases. In this analysis, we use purchase quantity as a proxy for monetary value since actual price data is not accessible.

b) RFM Scoring: By the usage of clustering analysis the RFM score combines three metrics, following the data mining methodology suggested by Wei et al. [3](2010). Each component is segmented into 4 clusters (1–4), where higher clusters representing better customer behavior. The elbow method is used to determine the optimal number of clusters for each metric (Recency, Frequency, and Monetary). Figure 2 illustrates the relationship between the number of clusters (k) and inertia for recency, where the "elbow point" at k=4 indicates the ideal number of clusters. The final RFM score is computed as:

$$\text{RFM Score} = (4 - \text{Recency}) + \text{Frequency} + \text{Monetary}$$

4) Customer-Product Interaction Features: The following features capture the relationship between customers and products:

a) Customer-Centric Metrics:

- Total unique products purchased
- Total purchase count
- Average purchase frequency
- Customer-product purchase ratio

b) Product-Centric Metrics:

- Total unique customers per product
- Total purchase count per product

5) Time Window Analysis: Rolling window features capture recent behavior patterns. Features calculated for windows of 7 days, 14 days, 30 days, and 90 days:

- Purchase counts
- Total quantities
- Purchase frequencies

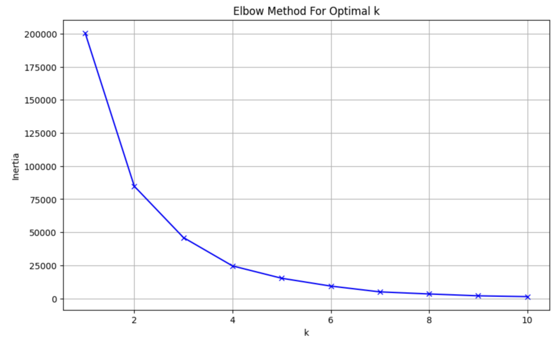

Fig. 2. Inertia by number of clusters for recency

D. Label Creation

The time between consecutive purchases for each customer-product pair was analyzed in order to create the target variable. For each transaction, we analyzed the time difference to the next purchase and assigned labels accordingly. With this method, the continuous time series data is coverted into a discrete classification problem, with each label representing a specific repurchase time window. The prediction labels were assigned according to the following time brackets:

- Label 1: Repurchase within 1-7 days
- Label 2: Repurchase within 8-14 days
- Label 3: Repurchase within 15-21 days
- Label 4: Repurchase within 22-28 days
- Label 0: No repurchase within 28 days

E. Data Filtering Strategy

Due to the competition's scoring mechanism, where correct repurchase predictions (Labels 1-4) earn both a base point and three bonus points for correct week prediction, while Label 0 predictions earn only a base point, we implemented a strategic filtering approach. We removed transactions with Label 0 from the training set to optimize our model's performance toward predicting repurchase weeks. This decision was made to focus the model's learning on patterns that lead to repurchases, thereby maximizing potential points in the competition's scoring system.

IV. Model Development

A. Model Architecture

1) XGBoost: XGBoost, a highly efficient gradient boosting framework, was utilized for multi-class classification in this project. Its ability to handle imbalanced datasets, support categorical features, and deliver high predictive accuracy made it an ideal choice.

2) CatBoost: CatBoost, another gradient boosting framework specifically designed to handle categorical features, was also employed. Its native support for categorical data reduces the need for extensive preprocessing, such as one-hot encoding, making it both efficient and accurate. Additionally, CatBoost is less prone to overfitting and achieves fast training times while maintaining high accuracy, making it a strong contender for multi-class classification tasks in our project.

3) Random Forests: Random Forests, an ensemble learning method based on decision trees, was also explored for this project. By building multiple decision trees and averaging their outputs, Random Forests provide a robust and interpretable model that is resistant to overfitting. This algorithm was particularly useful as a benchmark due to its simplicity and ability to capture nonlinear relationships in the data. Despite its limitations in handling categorical features without preprocessing, Random Forests provided valuable knowledge into feature importance and baseline performance.

B. Training Process

1) Training Methodology: To ensure temporal integrity in the dataset, a time-series cross-validation (CV) approach was adopted with 7 folds. Each fold maintained the chronological order of the data to prevent information leakage between training and validation sets.

The training steps included:

- Splitting the dataset into training and validation sets for each fold.
- Encoding categorical features numerically to enable compatibility with the XGBoost model.
- Training the model on the training data and evaluating it on the validation data.

2) Hyperparameter Tuning: The model's hyperparameters were fine-tuned based on performance metrics such as macro F1-score and custom score. Parameters such as learning rate, max depth, and subsample ratio were adjusted iteratively to optimize model performance. Grid search and manual tuning techniques were employed to identify the optimal configuration.

3) Validation Strategy: The cross-validation strategy was designed to assess the model's robustness across different time periods. Metrics such as macro F1-score and a custom score were computed for each fold, providing idea into the model's predictive capabilities. Validation performance guided further refinement of the model's parameters and feature engineering strategies.

V. Results and Discussion

A. Evaluation Metrics

To evaluate model performance, we used the following metrics:

1) F1 Score: The F1 score, a harmonic mean of precision and recall, was used to balance the assessment of false positives and false negatives. For multi-class classification, the macro-average F1 score was calculated:

$$F1_{\text{macro}} = \frac{1}{N}\sum_{i=1}^{N}\frac{2\cdot \text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}$$

This metric is ideal for imbalanced datasets, ensuring equal weight for all classes.

2) Custom Score: A custom score was designed to align with competition rules, rewarding correct repurchase predictions (Labels 1–4) with 1 base point and an additional 3 bonus points for accurate week predictions. Correct Label 0 predictions received only 1 base point. The custom score formula:

$$\text{Custom Score} = \sum_{i=1}^{N}(\text{Base Points}_i + \text{Bonus Points}_i)$$

This metric prioritizes precise week predictions for Labels 1–4.

B. Model Performances

We did nearly 30 submission for this competition which one of the highest if not the highest.

XGBoost emerged as the best-performing model, achieving a final score of 7170 on the complete dataset. Its ability to incorporate a wide range of features and optimize their importance through tuning contributed significantly to its superior results.

The Catboost model ignored most of the features we created, as a result it performed worse than XGBoost with 6800 at most.

The Random Forest model was implemented during the early stages of the project but struggled with the dominance of zero inputs in the dataset, resulting failure of getting anything useful.

VI. Conclusion

Our predictive analytics model faced challenges due to the imbalanced dataset and the lack of clear purchase motives, which impacted its performance across all approaches. Despite this, the project provided valuable insights into feature engineering, validation strategies, and problem-solving techniques. By addressing data sparsity through innovative methods like product grouping and focusing on customer behavior patterns, we developed a solid foundation for future applications. These methods can be scaled to larger or more diverse datasets to optimize inventory management and supply chain operations, showing the potential of predictive analytics in e-commerce.

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